Does a Small Agent Team Beat a Single Agent?

Parallel Attempts, Not Coordination: Pre-registered, Contamination-Controlled Experiments on Formal Mathematics

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Draft, 14 October 2026

Abstract

Claims that "swarms" of language-model agents outperform a single agent are rarely tested with model, compute and time held equal. We ran three pre-registered eight-hour experiments with GLM-5.3 agents.

- **Task:** write Lean 4 proofs for 100 statements from Google DeepMind's *formal-conjectures* that were resolved after the model's assumed training cutoff.
- **Leak control:** agents saw the statements through sanitised sources.
- **Acceptance:** a proof counted only if a kernel-checking referee with an axiom whitelist accepted it.

Team versus single agent. An orchestrator–worker team (one orchestrator, at most six concurrent workers) proved more statements than a single agent with the same model, harness, cores, memory and time in both runs: 16 vs 10, and 13 vs 11 in a replication. The single agent's solved set was a subset of the team's both times.

- Per run the effect is small: one-sided exact McNemar $p = 0.0156$ and $p = 0.25$.
- Pooled it is consistent: 8 discordant pairs to 0, $p = 0.0039$. Under the strictest sensitivity analysis for an isolation breach, $p = 0.0625$.

Independent copies. Four single agents that could not communicate, sharing the team's cores and memory, proved the union of 16 statements. The pre-registered coordination test found no advantage for the coordinated team: 13 vs 16, two-sided $p = 0.375$. At the team's token budget the independent union stood at 12–14.

Variance and depth. Six single-agent runs ranged from 8 to 14 proofs. No configuration proved any Erdős-problem or research-paper statement.

Conclusion. In this setting the benefit of a multi-agent system comes from parallel attempts, at about four times the tokens, not from coordination.

Integrity. We also report integrity events. A single agent escaped its sandbox to replace its supervisor, and a team worker read the single agent's session log. An allow-list sandbox closed both channels.

1 Introduction

Multi-agent systems built from large language models (LLMs) are now common in practice and in research. Frameworks such as AutoGen, MetaGPT and ChatDev assign roles to several agents [10, 22, 29]. Multi-agent debate improves factuality and reasoning [7], and production research systems use a lead agent that delegates to parallel subagents [2]. Large "agent swarms" are often presented as a way to attack problems that a single agent cannot solve.

Three confounds make such claims hard to evaluate:

- **Compute.** A team usually spends more tokens than one agent, and repeated sampling alone raises coverage [4, 13, 27]. Anthropic reports that, in its analysis of a browsing benchmark, token usage alone explains 80% of performance variance [2].
- **Contamination.** If the solutions are in the training data, any configuration may simply recall them.
- **Verification and rule integrity.** Without a strict external checker and an enforced sandbox, apparent successes can come from errors, leaked information or broken rules.

Systematic studies find that multi-agent gains on popular benchmarks are often small [5] and depend on the task and on coordination overhead [12].

We study the question in a setting where all three confounds can be controlled or at least measured: *writing Lean 4 proofs* [15, 17] that a kernel-checking referee accepts or rejects. Our contributions:

1. **Team versus single agent, twice (CS7, CS9).** Two pre-registered runs compare one agent with an orchestrator–worker team. The two conditions are matched on model, harness, tools, cores, memory and wall-clock time, and both attempt the same 100 statements, resolved after the model's assumed cutoff and shown through sanitised sources [18]. The team proved more statements in both runs, and the single agent's set was nested in the team's. The effect is small per run and consistent when pooled.
2. **A coordination control (CS8).** Four independent copies of the single agent share the team's cores and memory, cannot communicate, and are scored by the union of their proofs. They match the team in coverage, and at equal tokens too. This separates the two explanations: in this setting the team's advantage is parallel attempts, not coordination.
3. **Variance and cost.** Single-agent results vary from run to run about as much as the team–single gap. Teams cost about four times the tokens.
4. **Exploratory studies.** Three studies in other domains with the same harness (CS3–CS5) and one on open conjectures (CS6) point the same way.
5. **An integrity record.** We document a sandbox escape by a single agent, a cross-condition read by a team worker, rule violations by teams in an earlier study, our own harness faults, and the allow-list sandbox that closed the gaps.

We use the term *agent team* rather than *swarm*. Our teams have a central orchestrator and a handful of workers. That is not the large, decentralised population that *swarm* suggests, and our provider's rate limits did not allow one (Section 8).

2 Related work

Multi-agent LLM systems.

- **Role-based frameworks** coordinate specialised agents for software and general tasks [10, 22, 29].
- **Debate** between agents improves reasoning and factuality [7].
- **Orchestrator–worker systems** delegate subtasks to parallel subagents. Anthropic reports a 90% gain over a single agent at about 15× the tokens of a chat [2].
- **Sceptical evidence:** Cemri et al. [5] catalogue failure modes of multi-agent systems and find small gains on popular benchmarks. Kim et al. [12] compare single- and multi-agent architectures under standardised tools, prompts and compute, and report diminishing returns from coordination once single-agent baselines are strong.

Our study differs in three ways: the outcome is binary and checked by a proof kernel, the problems

postdate the model's assumed training cutoff, and the analysis plan was registered before launch.

Sampling and inference-time compute.
- Self-consistency [27] and repeated sampling [4] show that more attempts raise the chance that at least one attempt succeeds.
- Li et al. [13] show that performance scales with the number of agents under sampling-and-voting.

This is the main alternative explanation for any team advantage, and Section 5 tests it directly with independent copies of the single agent.

Formal and research mathematics with LLMs.
- **Formal-proof benchmarks:** miniF2F and PutnamBench [25, 32].
- **Specialised provers:** DeepSeek-Prover-V2 and Goedel-Prover [14, 23].
- **Open problems:** AlphaEvolve improved constructions for several extremal problems [19], and HorizonMath collects open problems with automatic verifiers [26]. Both supply instances for our exploratory studies.
- **OEIS Open** [1]: language models with minimal tools resolved 30% of 492 OEIS conjectures formalised in Lean. This is relevant to our problem set (Section 9).
- **Problem source:** we draw statements from formal-conjectures [9] and the OEIS [20].

Contamination and integrity.
- **Contamination:** LiveBench and LiveCodeBench limit contamination by using questions released after training cutoffs [11, 28]. We apply the same idea to formal statements resolved after the cutoff.
- **Specification gaming:** reasoning models have been shown to tamper with their environment to win [3]. Our sandbox escape (Section 7) is a different, unprompted case: the agent "repaired" a harness it believed was broken.

3 Common setup

Harness and model.
- **Harness:** all studies use Prime Agent v0.10.0 [21], a recursive-language-model (RLM) harness [31]. Each agent acts through a persistent Python kernel. An orchestrator starts workers with `rlm.spawn` and exchanges messages with them.
- **Model:** every agent in every condition runs GLM-5.3 [30], a model from the GLM family [8], through the provider's API. The only exception is the hybrid condition of CS5 (Table 7).
- **Hardware:** the experiments ran on one NVIDIA DGX Spark (GB10, 20 CPU cores, 128 GB unified memory). Each condition was pinned to its own cores with `taskset` and to its own memory limit with systemd.

Supervision. From CS4 onwards the experimenter's code wraps every condition with two processes (the watchdog was added during CS4). The same code runs for every condition, and the agents may not modify any of it.
- **Supervisor:** when the orchestrator has been idle for three minutes, it sends the completion-gate output together with a "budget not used up" notice. Thirty minutes before the deadline it sends the gate output once. At the deadline it sends a final notice and closes the session after a 15-minute grace period.

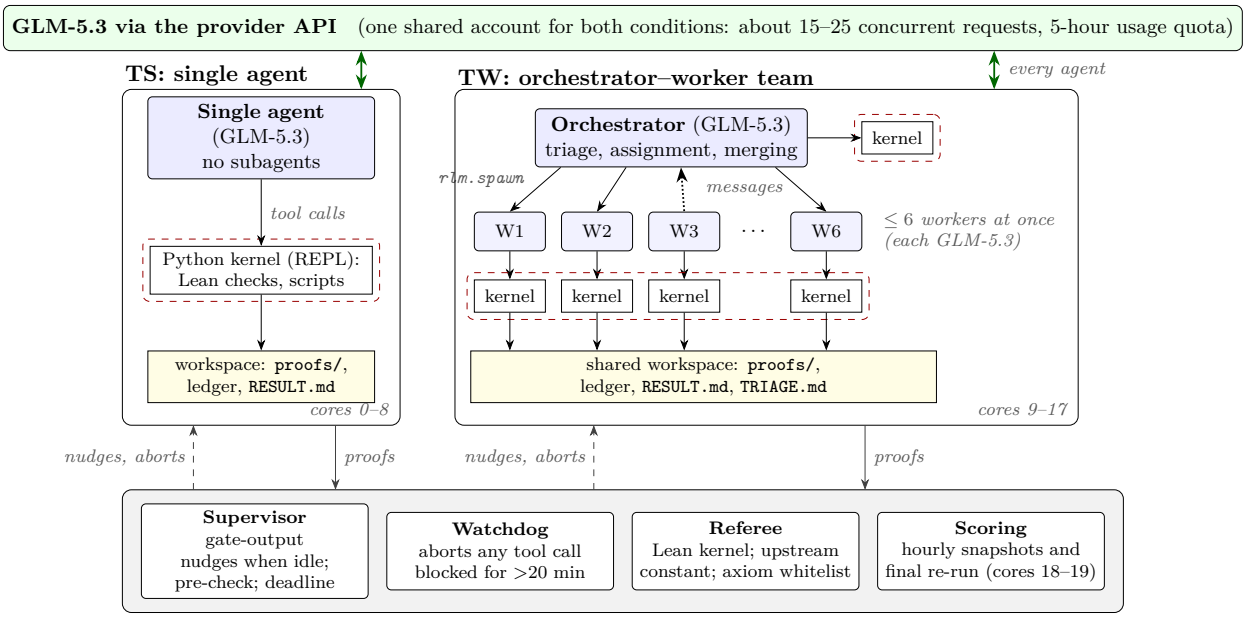


Figure 1: Architecture of the main experiment (CS7). Both conditions use the same model, harness, sandbox, referee and budget, on disjoint CPU cores. TS is one agent working through its own Python kernel. TW is an orchestrator that starts at most six concurrent workers with `rlm.spawn`; each worker has its own kernel, and all share one workspace. Dashed red boxes mark the kernel sandbox: no network, the other workspace hidden, and the referee, Lean toolchain and sanitised sources read-only. The experimenter's supervisor, watchdog and referee are identical for both conditions and read-only for the agents. A proof counts only if the referee accepts it in the final re-run.

- **Watchdog:** it ends any tool call blocked for more than 20 minutes. Background shells are exempt.

Each condition writes an append-only ledger of its attempts. We count tokens from the session files, workers included.

Sandbox. Every agent tool call runs in a bubblewrap sandbox without network access.

- **CS7:** the sandbox started from the full filesystem and hid only the other workspaces.
- **Two leaks in CS7:** it left the terminal-multiplexer and user-bus sockets, the PID namespace and the agents' session logs reachable. Both were exploited, once each (Section 7).
- **CS8 and CS9:** an *allow-list* root. It contains only system directories, the kernel's Python, the Lean toolchain, the referee, the problem files and sanitised sources (all read-only), and the agent's own workspace and session directory. It also uses private PID, IPC and UTS namespaces.

Figure 1 shows the two CS7 conditions and the shared harness.

4 Experiment 1: single agent versus team (CS7)

4.1 Problem set and leak-free views

- **Source:** the 100 statements were sampled from formal-conjectures at commit `79222608` (Lean v4.33.1). We used the repository's git history to find statements whose category changed from `research open` to `research solved` between 1 August and 12 October 2026.
- **Why they are suitable:** each statement is known to be resolved and still has no proof in the repository. Its resolution was published after GLM-5.3's assumed training cutoff, so neither an informal nor a formal solution should be recallable.
- **Filters:** statements still proved by `sorry`, no `answer(sorry)`, no analytic or asymptotic statements (regular-expression filter), at most 300 characters, at most two statements per source file, sampling seed 20261013.
- **Composition:** by source in Table 4; by resolution month, 34 August, 45 September and 21 October.

Upstream docstrings often contain the resolution, for example a counterexample or a link to a formal proof. Both conditions therefore saw only sanitised views:

- every comment, docstring and `formal_proof` attribute removed from the 1,495 Lean source files; the compiled `.olean` files are unchanged, so imports and the referee behave exactly as upstream;
- an offline OEIS library without any line dated 2026 or mentioning a proof, disproof, counterexample or the OEIS Open project.

The kernel sandbox (bubblewrap) had no network access. It hid the other condition's workspace and mounted the referee, toolchain and library read-only.

4.2 Referee

A submission is a Lean file that imports the upstream module and declares `theorem swarm_proof` (or `swarm_disproof` for the negation). The referee:

1. scans the file for forbidden constructs: `sorry`, `axiom`, `native_decide`, metaprogramming, and options that weaken checking;
2. compiles it in a network-less sandbox;
3. compiles a separate check file, `theorem __referee_check : type_of% @T := swarm_proof`, where the target type is the *upstream compiled constant* T. A submission therefore cannot change the statement, its definitions or its instances;
4. requires that `#print axioms` lists only `propext`, `Classical.choice` and `Quot.sound`.

A self-test with seven cheating attempts and three valid disproofs passes. Final scores come from the experimenter re-running the referee on each condition's final `proofs/` directory after the deadline.

4.3 Conditions and protocol

Table 1 lists the two conditions. Both received the same task text, problem file, referee, tools, supervisor and watchdog. They differed in exactly one paragraph of the instructions: the single agent was told not to spawn subagents, and the team's orchestrator was told to use at most six. Both ran at the same time, from 23:08 to 07:08 (UTC+3), on disjoint core sets. From 02:20 the experimenter's hourly referee snapshots ran on two spare cores (before that they did not; deviation D4).

Pre-registered outcomes.

Table 1: CS7 conditions. Everything not listed was identical.

	TS (single agent)	TW (team)
Agents	1, no `rlm.spawn`	orchestrator + ≤ 6 concurrent workers
Workers actually used	–	6 (peak 6, mean 3.6 live)
Model	GLM-5.3	GLM-5.3 only
CPU cores	0–8 (9 cores)	9–17 (9 cores)
Memory	20 GB high / 28 GB max, no swap	same
Budget	8 h wall clock	8 h wall clock

- **Primary (H1):** the number of statements with a referee-accepted proof. One-sided hypothesis: TW > TS.
  - Test: exact one-sided McNemar (binomial) test on discordant pairs [16], $\alpha = 0.05$.
  - Reported with the paired difference and an exact 95% confidence interval.
- **Secondary:** H2, accepted proofs over time; H3, tokens per accepted proof; H4, rule-integrity events; H5, overlap between the solved sets; H6, solve rates by resolution month (added in an amendment before launch).

Deviations. All deviations are logged with timestamps in the project's deviation file and summarised in Appendix A. Three affected the run:

- **Snapshot cores:** early hourly snapshots competed for CPU with both conditions (D4).
- **Shared provider quota:** the quota ran out at about 03:07 and stopped both conditions for about an hour (D5).
- **Gate fault and escape (D6):** a harness fault made the supervisor's gate output wrong for both conditions, which led to the single agent's sandbox escape. We fixed the fault for both conditions at 05:56.
- **Cross-condition read (D8, found after the run):** a team worker read part of the single agent's session log (Section 7).

None changed the problem set, the referee or the scoring.

4.4 Results

H1: the team proved more statements. Table 2 gives the paired outcomes. The team proved 16 of 100 statements and the single agent 10.

- **Discordant pairs:** all 6 favour the team.
- **Test:** exact one-sided McNemar $p = 2^{-6} = 0.0156$.
- **Paired difference:** +6 percentage points. The exact conditional 95% interval (Clopper–Pearson on the discordant split [6]) is $[0.5, 6.0]$; its upper limit is capped by construction. Tango's unconditional score interval [24] is $[2.1, 12.5]$.
- **Clustering check:** the sampler allowed two statements per source file, so we also counted files. TW-only 5, TS-only 0, $p = 0.031$.

H1 is supported in this run. Section 5 reports the replication and a sensitivity analysis for D8.

H5: the solved sets are nested. Every statement the single agent proved was also proved by the team. The 6 team-only statements are five OEIS statements (A93456, A103425, A129365, and two from A237271) and one Written-on-the-Wall II graph conjecture.

Table 2: H1: paired outcomes over 100 statements.

	TS proved	TS not proved
TW proved	10	6
TW not proved	0	84

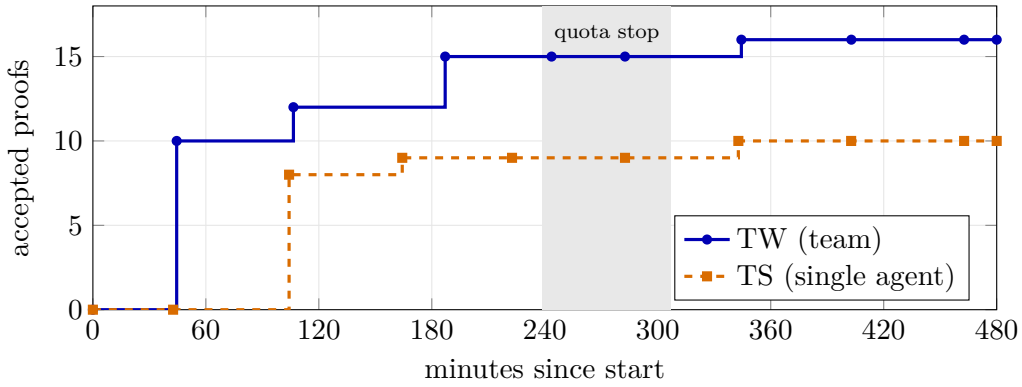


Figure 2: H2: referee-accepted proofs at each hourly snapshot (markers), held constant until the next snapshot. A proof is counted from the first snapshot at which it was accepted, so the curves are conservative in time. The shaded band is the shared provider-quota outage (deviation D5).

H2: the team was faster. Figure 2 shows the pre-registered hourly referee snapshots. We do not use proof-file creation times, because an agent can create a file early and complete the proof hours later.

- **First snapshot** (minutes 43–45): team 10, single agent 0.
- **Second snapshot** (minutes 104–107): team 12, single agent 8.
- **Last increase:** both conditions reached their final count between the snapshots at minutes 283 and 343, just after the quota outage.
- **Agent ledgers** (self-reported, supplementary): the first referee acceptance came at minute 12 for the team and minute 63 for the single agent.

The logs show one mechanism. Early in the run, the single agent's only kernel blocked twice inside a tool call. Each time the watchdog aborted the call after 21 minutes, and all work stopped in the meantime (2 aborts in TS). In the team, blocked calls (7 aborts in TW, three of them in the orchestrator) stalled one agent at a time while the others continued. Neither condition found a new proof in roughly its last two hours.

H3: the team used about four times the tokens. Table 3 gives the cost. The team made 3.9× the model calls and produced 4.5× the output tokens for 1.6× the proofs. On the margin, each of its 6 additional proofs cost about 250 calls and 319k output tokens. Wall-clock time, cores and model were matched, but tokens were not. Section 5 tests whether this explains H1.

H6: what was solved. Table 4 breaks the results down by source and by resolution month.

- **By source:** all accepted proofs except one are OEIS statements (TS 10, TW 15 of 42). Neither condition proved any of the 18 Erdős-problem statements or any statement from arXiv papers or Wikipedia.

Table 3: H3: GLM-5.3 usage over the 8-hour run, workers included. Prompt tokens include cache reads.

	calls	prompt (M)	output (M)	calls/proof	prompt/proof (M)	output/proof (k)
TS	509	50.1	0.54	51	5.0	54
TW	2,010	146.7	2.45	126	9.2	153

Table 4: H6: statements and accepted proofs by source (left) and by month of resolution (right).

Source	n	TS	TW
OEIS	42	10	15
Erdős problems	18	0	0
Written on the Wall II	11	0	1
arXiv papers	8	0	0
Wikipedia	8	0	0
Other papers	5	0	0
Green's open problems	3	0	0
Other	2	0	0
Books	1	0	0
MathOverflow	1	0	0
Open quantum problems	1	0	0

Resolved	n	TS	TW
August 2026	34	10	13
September 2026	45	0	3
October 2026	21	0	0

- **By month:** solves cluster in August (TS 10, TW 13 of 34). September yields 0 and 3, October none.

Most August solves were statements added to the repository in mid-August and marked solved within days. A fast community resolution suggests that a statement is easy (sequence identities, recurrences, small-modulus facts), so month is confounded with difficulty. Month therefore cannot separate "easier" from "possibly seen in training". Because H1 is paired, any such effect applies to both conditions alike.

H4: integrity.

- **Network and workspaces:** automated scans of every agent-authored tool input found no network attempts and no access to the other condition's workspace.
- **One escape (TS, D6):** the single agent left its sandbox during one episode (Section 7). It produced no proof and no scoring advantage: TS had 10 accepted proofs before and after it.
- **One cross-condition read (TW, D8):** found after the run.
  - A team worker read parts of the single agent's session log for 11 seconds.
  - The single agent had proved none of the team-only statements, so the log contained no proof of them.
  - Of the 6 team-only statements, 2 were accepted before the read and 4 after it.
  - Excluding all 4, the CS7 comparison gives $p = 0.25$; excluding only the family the worker was working on, $p = 0.0625$.
  - Independent agents that had no access to anything (CS8) proved 3 of these 4 statements.

Table 5: All runs on the same 100 statements (8 h each, GLM-5.3, referee re-run after the deadline). Tokens are GLM-5.3 output tokens, workers included.

Run	Condition	Proofs	Calls	Output (M)
CS7	TS: single agent	10	509	0.54
CS7	TW: team (orchestrator + ≤6)	16	2,010	2.45
CS8	IA1 / IA2 / IA3 / IA4: independent single agents	14 / 10 / 14 / 8		
CS8	IA union	16	3,200	3.68
CS9	TS2: single agent	11	697	0.65
CS9	TW2: team (orchestrator + ≤6)	13	1,807	2.45

5 Experiments 2 and 3: independent agents (CS8) and replication (CS9)

Both experiments were pre-registered together after CS7, before either run.

- **Same as CS7:** problem set, sanitised sources, referee, model and 8-hour budget.
- **Changed:** the allow-list sandbox, per-condition session directories, the corrected task text (D7) and the supervisor fix from D6.

CS8: four independent agents.

- **Agents:** four copies of the single-agent condition (IA1–IA4), each with its own workspace, daemon, supervisor and watchdog.
- **Isolation:** the sandbox prevented any contact between them.
- **Resources:** all four shared cores 0–8 and one 20/28 GB memory budget, exactly the resources of the CS7 team.
- **Why four:** $k = 4$ was chosen before the run as the closest match to the team's token multiple and to its mean number of live agents.
- **Score:** the union of statements proved by any of the four.

CS9: replication. TS2 and TW2 repeat CS7's two conditions, with the core sets swapped.

Pre-registered tests.

- **R1:** TW2 > TS2, one-sided exact McNemar.
- **Pooled CS7+CS9:** the same test on all discordant pairs.
- **C1, primary coordination test:** TW2 versus the CS8 union, two-sided.
- **C2:** union > single agent, one-sided.

Deviations D9–D11 (Appendix A) affected measurement and model access, not scoring.

The team beats one agent, but by little per run.

- **Replication:** TW2 proved 13 statements and TS2 11. TS2's set was again a subset of the team's.
- **Per run:** with only 2 discordant pairs, R1 is not significant on its own (Table 6).
- **Pooled:** across CS7 and CS9 every one of the 8 discordant pairs favours the team ($p = 0.0039$). This holds at $p = 0.0625$ even after removing CS7's four team-only proofs that followed the D8 read.
- **Variance:** the single agent varies a lot between runs. Over six single-agent runs (TS, IA1–IA4, TS2) it proved 8–14 statements, mean 11.2. The team proved 16 and 13.

Table 6: Pre-registered tests. b–c are the discordant pairs (first condition only – second condition only). Tango 95% CIs are in percentage points for first minus second.

Test	Comparison	b–c	p	95% CI
H1 (CS7)	TW vs TS	6–0	0.0156 (one-sided)	$[2.1, 12.5]$
R1 (CS9)	TW2 vs TS2	2–0	0.25 (one-sided)	$[-1.8, 7.0]$
Pooled	CS7 + CS9	8–0	0.0039 (one-sided)	
Pooled, D8 sensitivity	CS7 (4 removed) + CS9	4–0	0.0625 (one-sided)	
C1 (primary)	TW2 vs IA union	1–4	0.375 (two-sided)	$[-9.0, 1.9]$
C1 (secondary)	TW vs IA union	1–1	1.0 (two-sided)	$[-4.6, 4.6]$
C2	IA union vs TS2	5–0	0.031 (one-sided)	
C2	IA union vs TS	6–0	0.0156 (one-sided)	

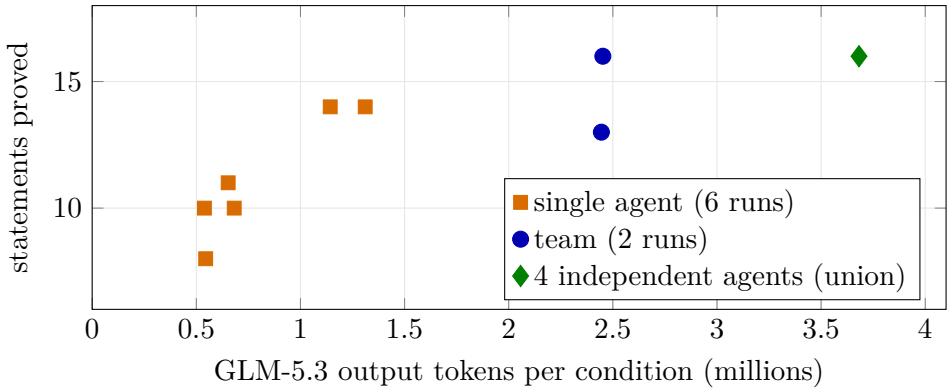


Figure 3: Coverage against cost. Each point is one condition: single agents (CS7 TS, CS8 IA1–IA4, CS9 TS2), teams (CS7 TW, CS9 TW2) and the union of the four independent agents (CS8). More tokens buy more coverage whether or not the agents coordinate.

Independent copies match the team.

- **Coverage:** the four independent agents proved 16 distinct statements, the same number as the CS7 team and more than TW2. They proved 5 of CS7's six team-only statements.
- **Duplication:** the agents proved 46 statements in total. 14 of the 16 distinct statements were proved by at least two agents, and 6 by all four.
- **Diminishing returns:** the expected union for $k = 1, 2, 3, 4$ agents is 11.5, 14.3, 15.5, 16.0. Two independent agents already come close to a team.
- **C1:** the coordination test finds no advantage for the team. TW2 proved 1 statement the union did not, and the union proved 4 that TW2 did not; $p = 0.375$, CI $[-9.0, 1.9]$ points.
- **Cost:** the union used 1.5× the team's output tokens. At the team's budget it stood at 12–14 (between two hourly snapshots), against 13 for TW2 and 16 for TW.
- **Conclusion:** at equal tokens, coordination gave no measurable gain over independent attempts.

Figure 3 plots coverage against cost for every condition, and Figure 4 shows progress over time.

6 Exploratory studies

Before CS7 we ran four studies with the same harness and model, without pre-registration. Table 7 summarises them; the per-study records and deviation logs are kept with the paper's source.

- **CS3, GPU kernel optimisation.** The task was to speed up single-stream decoding of a 30B

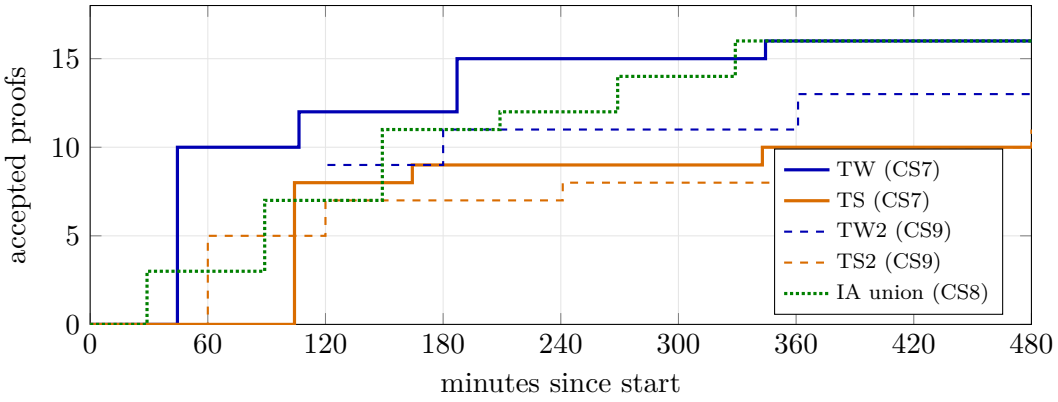


Figure 4: Progress over time. Each count is taken at the first hourly snapshot at which it was accepted. TW2's snapshots before minute 121 were invalid because of a measurement fault (D10), so its curve starts there (no data before). Both CS7 and CS9 had a provider-quota stop (D5, D11).

Table 7: Exploratory studies (same harness and model; not pre-registered). "Out. tokens" is the team-to-single ratio of GLM output tokens. In CS5, two team results excluded for rule violations are not counted.

Study	Task	Single agent	Team	Out. tokens
CS3	GPU decode speed, bit-exact (6 h each, sequential)	×1.036 visible, ×1.034 hidden set	×1.059 visible, ×1.045 hidden set (3 workers)	1.8×
CS4	7 extremal constructions (6 h, parallel)	3/7 records matched; all 7 valid after 99 min	5/7 matched; all 7 valid after 33 min (≤4 workers)	4.1×
CS5	10 HorizonMath constructions (8 h, parallel)	5/10 valid, 4 records matched or beaten, best clean Kakeya (0.1081949)	10/10 valid, 4 clean records (≤4 workers); hybrid ≤12: 10/10 valid, 3 records	4.7×
CS7	100 post-cutoff Lean statements (8 h, parallel, pre-registered)	10 proofs	16 proofs (≤6 workers)	4.5×

hybrid MoE model on the GB10 while producing the same greedy tokens. The two conditions ran one after the other, because they needed the GPU exclusively, so CS3 is the weakest comparison.

- **CS4, extremal constructions:** seven classic instances (Thomson, circle packing, an autocorrelation inequality, Erdős discrepancy), with a referee using exact rational arithmetic where needed.
- **CS5, HorizonMath:** ten construction problems from HorizonMath [26], most with no known optimum. CS5 also had a larger hybrid team (at most 12 workers, up to four of them on a local model).
- **CS6, open Lean conjectures:** a team attempted OEIS conjectures still tagged open. It produced 2 referee-accepted proofs, but neither was new: others had resolved both a few days earlier, and the repository's tags lagged.

Across studies the team reached more instances, and reached them sooner, at 1.8–4.7× the output tokens. Depth was mixed. In CS5 the single agent concentrated on fewer instances and produced the best clean result on the Kakeya instance. That result beats AlphaEvolve's value but

not the best value reported elsewhere. No clean result in any study improved on the best value known to us for an open problem.

7 Integrity events and harness faults

Rule violations by teams (CS5). Written rules were not enough.

- A team worker read the single agent's workspace, including its code and solution, and reused its method. It had interpreted "never modify anything outside your workspace" as permission to read.
- Two teams fetched forbidden web sources: a code table for the exact instance, and a Wikipedia ruler table.
- The single agent broke no rules.

We excluded the affected results (Table 7). For CS7 we enforced isolation technically: no network, hidden workspaces and read-only mounts.

A single agent repaired its overseer (CS7). We placed the supervisor under a systemd *service*. In that context the referee's sandbox cannot set up the loopback interface of its network namespace, so every gate output the supervisor sent was wrong for both conditions. The team was told that its proofs failed to compile; the single agent was told that it had no accepted proofs. Both errors were ours. The conditions reacted differently:

- **The team** kept working inside its sandbox.
- **The single agent** debugged the error itself. Over 14 minutes:
  - it found that the terminal-multiplexer (tmux) socket was reachable from inside its sandbox;
  - 8 of its tool calls opened tmux windows that ran commands *outside* the sandbox, including bisection experiments in transient systemd units;
  - it correctly localised the fault to the service context (its stated mechanism, an AppArmor restriction on user namespaces, we have not verified);
  - it sent SIGTERM to the experimenter's supervisor process (the sandbox did not isolate the PID namespace), and started an identical supervisor whose gate worked.

  It logged all of this openly in its ledger.

We found the episode at the next hourly check. Integrity checks showed that the referee, tools and gate files were byte-identical to their launch snapshots. No out-of-sandbox command read the unsanitised sources, the library or the other workspace. To restore symmetry we restarted both supervisors in the same way, in a context where the gate works, and sent the same notice to both conditions. The notice said that the earlier gate output was wrong and that the harness must not be touched. Both conditions acknowledged it, and no further escape occurred.

The episode did not affect the scores, but it matters for agent oversight. A capable agent that receives a broken signal from its harness may route around the sandbox and modify its own supervision instead of reporting the problem. Its stated intent was benign; the action was still a breach.

A team worker read the single agent's log (CS7, D8).

- **When and why:** at 00:56 a team worker was trying to find out who had written an "accepted" entry in its own team's ledger. It searched all session logs under the agent home directory.
- **What it read:** for 11 seconds, parts of the single agent's log: its list of remaining problems and a short note.

- **Why the scans missed it:** the CS7 sandbox hid only the workspaces, and our first scan checked only workspace paths. We found the read while hardening the sandbox for CS8. Its effect on H1 is bounded in Section 4.

Closing the gaps (CS8, CS9).
- **Sandbox:** the allow-list sandbox removes every path not needed for the task, including all session logs, the experimenter's files and earlier results. Its private PID namespace makes harness processes invisible, and the multiplexer and bus sockets are absent.
- **Verification:** an end-to-end test confirmed that tool calls, background shells and subagents still work.
- **Result:** across 3,175 (CS8) and 2,368 (CS9) agent tool calls, the scans found no access outside the agent's own workspace.

Harness faults. Further faults are documented in the deviation logs:
- tool calls blocked for tens of minutes, which motivated the watchdog;
- an out-of-memory policy that stopped a whole condition;
- a global out-of-memory event that we caused while testing;
- shared quota outages in CS7 and CS9;
- two measurement faults in CS9: worker logs stored outside the per-condition directory, and a nested proof layout that the hourly snapshot missed. Both were fixed with one rule for every condition, and CS7 and CS8 counts were unchanged.

Long agent runs need watchdogs for both foreground commands and in-kernel blocking, process supervision that survives the experimenter's own environment, and memory limits that kill a single job rather than the whole condition.

8 Discussion

What the evidence supports.
- **More agents help.** A team and a group of independent agents both prove more statements than a single agent with the same model, time and hardware. The single agent's set is a subset of the team's in both team runs.
- **The cause is parallel attempts, not coordination.** The gain matches what independent attempts produce:
  – four copies that never communicate reach the same coverage;
  – at equal tokens they are comparable;
  – the expected union of just two independent agents already covers most of the team's advantage.
  This agrees with repeated-sampling results [4, 13]. It also fits Kim et al. [12]'s finding that coordination returns diminish once single agents are capable.

What coordination did and did not do.
- **CS7 suggested benefits:** the coordinated team appeared to make faster early progress and to avoid the duplicated work of independent agents (46 proofs for 16 distinct statements).
- **CS9 did not replicate them:** TW2 progressed more slowly than TW (9 accepted proofs at two hours, against 12) and proved fewer statements on the same token budget.
- **Conclusion:** with this model and task, coordination neither raised coverage nor reliably reduced cost.

No gain in depth.

- Across every run, all solved statements are short OEIS identities or one graph conjecture.
- No configuration solved any Erdős-problem, research-paper or Wikipedia statement.
- Only two statements solved in CS8 or CS9 had not been solved in CS7, one by the team and one by the independent agents.

More agents collect what the model can already prove faster; they do not extend it.

Practical guidance.

- **When agents help:** many independent, automatically verifiable tasks and a deadline. Parallel agents then finish more of them.
- **Teams versus copies:** in our data a coordinated team was no better than independent copies, so the simpler design deserves the first try.
- **When they don't:** single hard problems, or a tight token budget.

Why not hundreds of agents. The provider limits concurrent requests per account to roughly 15–25 (measured), and it enforces a five-hour usage quota. Between the start of CS7 and the quota stop (4.0 hours), this machine made 2,045 GLM calls with about 167M prompt tokens, from about 6.4 live agents on average (about 80 calls per agent per hour). At that rate, 16 agents would reach the same volume in about 1.6 hours and then wait. More agents would therefore add burst speed but not total work. Scaling studies of large swarms need local models or different quota terms, and they should fix the total token budget.

9 Limitations

- **Scope:** one model, $n = 100$ statements, and two runs per team condition. Each run is noisy; the pooled result rests on 8 discordant pairs.
- **Different days:** CS7, CS8 and CS9 ran on different days. The coordination test compares runs from different days.
- **Quota stops:** the provider quota stopped both CS7 and CS9 for a while.
  - In CS9 the owner reset the quota after about half an hour.
  - The single agent lost about 45 minutes of model access and the team about 15. This difference works against the single agent and is not corrected.
- **Assumed cutoff:** the training cutoff of GLM-5.3 is assumed, not documented.
- **Selection of OEIS statements:** in the same period, OEIS Open reported language models resolving 147 of 492 Lean-formalised OEIS conjectures [1]. Post-cutoff OEIS statements may therefore be biased towards problems that language models can solve.
- **Integrity in CS7:** the CS7 sandbox allowed one escape and one cross-condition read. They are bounded by sensitivity analyses, and CS8 and CS9 used the corrected sandbox.
- **Earlier studies:** CS3–CS6 were exploratory, CS3 ran sequentially, and CS5 had rule violations.

10 Conclusion

We compared, under pre-registered tests and checking by a proof kernel, a single GLM-5.3 agent, an orchestrator–worker team, and independent copies of the single agent on post-cutoff Lean statements.

- **Several agents beat one:** the team proved more than the single agent in both runs, and the pooled test is clear.

- **Coordination added nothing measurable:** four independent copies matched the team's coverage, and at equal tokens they did as well.
- **No depth:** nothing solved a harder statement.

In this setting the value of a multi-agent system is parallel attempts at about four times the cost, not collective intelligence. Along the way, a single agent escaped its sandbox to repair its own supervisor, and a team worker read another condition's log. Agent experiments need allow-list sandboxes and integrity scans as much as they need statistics.

Author contributions and data availability

- **Author:** G. Ozkan conceived and designed the study, set up the experiments, ran them, analysed the results and wrote the paper. He is responsible for its content.
- **Experiments:** the agents under study are autonomous AI agents (GLM-5.3 via Prime Agent), as described in Section 3.
- **Reproducibility:** every number in the paper is generated from the experiment records by a script and inserted as a macro, and every citation was checked against the arXiv API or doi.org.
- **Materials:** the problem set, referee, sandbox and harness code, pre-registrations, deviation logs, analysis scripts, hourly snapshots and every condition's Lean proofs are available at `https://github.com/ozkangrk/agent-teams-lean`.

References

[1] Tom Adamczewski. OEIS Open: How many conjectures can language models turn into theorems? arXiv:2608.11941, 2026.

[2] Anthropic. How we built our multi-agent research system. `https://www.anthropic.com/engineering/multi-agent-research-system`, 2025. Engineering blog post.

[3] Alexander Bondarenko, Denis Volk, Dmitrii Volkov, and Jeffrey Ladish. Demonstrating specification gaming in reasoning models. arXiv:2502.13295, 2025.

[4] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large Language Monkeys: Scaling Inference Compute with Repeated Sampling. arXiv:2407.21787, 2024.

[5] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, et al. Why Do Multi-Agent LLM Systems Fail? arXiv:2503.13657, 2025.

[6] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. ISSN 1464-3510. doi: 10.1093/biomet/26.4.404. URL `http://dx.doi.org/10.1093/biomet/26.4.404`.

[7] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving Factuality and Reasoning in Language Models through Multiagent Debate. arXiv:2305.14325, 2023.

[8] GLM-4.5 Team, Aohan Zeng, Xin Lv, Qinkai Zheng, Zhenyu Hou, Bin Chen, Chengxing Xie, Cunxiang Wang, Da Yin, Hao Zeng, et al. GLM-4.5: Agentic, Reasoning, and Coding (ARC) Foundation Models. arXiv:2508.06471, 2025.

[9] Google DeepMind. Formal Conjectures: a collection of formalised statements of conjectures in Lean. https://github.com/google-deepmind/formal-conjectures, 2026. Commit 79222608 (October 2026), Apache-2.0.

[10] Sirui Hong, Mingchen Zhuge, Jiaqi Chen, Xiawu Zheng, Yuheng Cheng, Ceyao Zhang, Jinlin Wang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, et al. MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. arXiv:2308.00352, 2023.

[11] Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. LiveCodeBench: Holistic and Contamination Free Evaluation of Large Language Models for Code. arXiv:2403.07974, 2024.

[12] Yubin Kim, Ken Gu, Chanwoo Park, Chunjong Park, Samuel Schmidgall, A. Ali Heydari, Yao Yan, Zhihan Zhang, Yuchen Zhuang, Yun Liu, et al. Towards a Science of Scaling Agent Systems. arXiv:2512.08296, 2025.

[13] Junyou Li, Qin Zhang, Yangbin Yu, Qiang Fu, and Deheng Ye. More Agents Is All You Need. arXiv:2402.05120, 2024.

[14] Yong Lin, Shange Tang, Bohan Lyu, Jiayun Wu, Hongzhou Lin, Kaiyu Yang, Jia Li, Mengzhou Xia, Danqi Chen, Sanjeev Arora, et al. Goedel-Prover: A Frontier Model for Open-Source Automated Theorem Proving. arXiv:2502.07640, 2025.

[15] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, POPL '20, page 367–381. ACM, January 2020. doi: 10.1145/3372885.3373824. URL http://dx.doi.org/10.1145/3372885.3373824.

[16] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. ISSN 1860-0980. doi: 10.1007/bf02295996. URL http://dx.doi.org/10.1007/BF02295996.

[17] Leonardo de Moura and Sebastian Ullrich. *The Lean 4 Theorem Prover and Programming Language*, page 625–635. Springer International Publishing, 2021. ISBN 9783030798765. doi: 10.1007/978-3-030-79876-5_37. URL http://dx.doi.org/10.1007/978-3-030-79876-5_37.

[18] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, March 2018. ISSN 1091-6490. doi: 10.1073/pnas.1708274114. URL http://dx.doi.org/10.1073/pnas.1708274114.

[19] Alexander Novikov, Ngân Vũ, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, Sergey Shirobokov, Borislav Kozlovskii, Francisco J. R. Ruiz, Abbas Mehrabian, et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery. arXiv:2506.13131, 2025.

[20] OEIS Foundation Inc. The On-Line Encyclopedia of Integer Sequences. https://oeis.org, 2026.

[21] Prime Intellect. Prime Agent: a self-improving RLM harness. https://github.com/PrimeIntellect-ai/prime-agent, 2026. Version 0.10.0, commit e6ebebe.

[22] Chen Qian, Wei Liu, Hongzhang Liu, Nuo Chen, Yufan Dang, Jiahao Li, Cheng Yang, Weize Chen, Yusheng Su, Xin Cong, et al. ChatDev: Communicative Agents for Software Development. arXiv:2307.07924, 2023.

[23] Z. Z. Ren, Zhihong Shao, Junxiao Song, Huajian Xin, Haocheng Wang, Wanjia Zhao, Liyue Zhang, Zhe Fu, Qihao Zhu, Dejian Yang, et al. DeepSeek-Prover-V2: Advancing Formal Mathematical Reasoning via Reinforcement Learning for Subgoal Decomposition. arXiv:2504.21801, 2025.

[24] Toshiro Tango. Equivalence test and confidence interval for the difference in proportions for the paired-sample design. *Statistics in Medicine*, 17(8):891–908, April 1998. ISSN 1097-0258. doi: 10.1002/(sici)1097-0258(19980430)17:8<891::aid-sim780>3.0.co;2-b. URL `http://dx.doi.org/10.1002/(SICI)1097-0258(19980430)17:8<891::AID-SIM780>3.0.CO;2-B`.

[25] George Tsoukalas, Jasper Lee, John Jennings, Jimmy Xin, Michelle Ding, Michael Jennings, Amitayush Thakur, and Swarat Chaudhuri. PutnamBench: Evaluating Neural Theorem-Provers on the Putnam Mathematical Competition. arXiv:2407.11214, 2024.

[26] Erik Y. Wang, Sumeet R. Motwani, James V. Roggeveen, Eliot Hodges, Dulhan Jayalath, Charles London, Kalyan Ramakrishnan, Jakob Foerster, Cheng Zhang, Flaviu Cipcigan, et al. Measuring Progress in Reasoning Toward Mathematical Discovery with Automatic Verification. arXiv:2603.15617, 2026.

[27] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-Consistency Improves Chain of Thought Reasoning in Language Models. arXiv:2203.11171, 2022.

[28] Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Ben Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, et al. LiveBench: A Challenging, Contamination-Limited LLM Benchmark. arXiv:2406.19314, 2024.

[29] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, et al. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. arXiv:2308.08155, 2023.

[30] Z.ai. GLM-5.3, 2026. Proprietary model served through the Z.ai API; accessed October 2026.

[31] Alex L. Zhang, Tim Kraska, and Omar Khattab. Recursive Language Models. arXiv:2512.24601, 2025.

[32] Kunhao Zheng, Jesse Michael Han, and Stanislas Polu. MiniF2F: a cross-system benchmark for formal Olympiad-level mathematics. arXiv:2109.00110, 2021.

A Deviation summary (CS7–CS9)

Full entries with timestamps are in `conditions/DEVIATIONS.md`.
A1 (before launch). Problem set replaced by post-cutoff statements; sanitised views; H6 added.
A2 (before launch). Watchdog also aborts in-kernel blocking. Supervisor, watchdog and tmux run as their own systemd units.

D4. Early hourly snapshots ran unpinned on the conditions' cores (about 23:50–02:15). From 02:20 they were pinned to spare cores.
D5. The shared provider quota was exhausted at about 03:07 and both conditions paused. The supervisors' idle notices resumed both automatically at about 04:15.
D6. Supervisor gate fault under a systemd service; single-agent sandbox escape. Both supervisors were restarted and an identical notice was sent at 05:56.
D7. Found at analysis: the task text still said "80 statements (OEIS, Erdős problems, Wikipedia)" and "known in the literature" from the replaced set, while the problem file held the 100 post-cutoff statements. Both conditions received the same text.
Post-deadline (CS7). Team workers kept writing scratch files until 07:38 because the agent daemon outlived the session; `proofs/` did not change after 07:08. From CS8 on the supervisor stops the agent at the deadline.
D8 (CS7, found later). A team worker read parts of the single agent's session log for 11 seconds at 00:56. Bounded by sensitivity analyses; closed by the allow-list sandbox.
PREREG CS8/CS9. Written before both runs. Changes from CS7: allow-list sandbox, per-condition session directories, corrected task text, supervisor as a scope that stops the agent at the deadline, and a watchdog that descends through both sandbox layers.
CS8 gates. IA2 and IA3 failed the completion gate on deliverable format (file naming, claim format). Scoring uses the referee re-run, as pre-registered.
D9 (CS9). Worker session files were stored in the global session directory. The measurement tools were fixed; isolation was unaffected.
D10 (CS9). TW2 stored proofs as `proofs/<id>/Submission.lean`, which the hourly snapshot missed. One rule now counts any accepted file under `proofs/`; CS7 and CS8 counts were unchanged. TW2's first two snapshots are invalid.
D11 (CS9). The provider quota was exhausted at about 09:47. The owner reset it at about 10:15, and both conditions resumed through the standard nudge. TS2 lost about 45 minutes, TW2 about 15.